

Professional Summary:

Experienced Data Engineer with 6 years of expertise in designing, evolving, and managing large-scale data infrastructures across AWS and Azure. Skilled in Python, SQL, Apache Spark, and Kafka to build scalable ETL pipelines, optimize real-time data processing, and integrate predictive analytics. Proven track record of improving data handling efficiency by 30% and accelerating analytical insight delivery by 25%. Adept at leading teams, mentoring engineers, and applying Agile principles to drive continuous data platform improvements. Hold a Master's degree in Computer and Information Science, with a strong foundation in data analytics and a commitment to advancing skills at the intersection of data engineering and AI.

Technical Skills:

Databases	Snowflake, MySQL, Oracle, Microsoft SQL Server, AWS Redshift, Amazon RDS, Teradata, PostgreSQL.
NoSQL Databases	MongoDB, Hadoop HBase, DynamoDB, and Apache Cassandra.
Programming Languages	Python, Scala, SQL, Java
Cloud Platforms	AWS, Azure
Querying Languages	SQL, NoSQL, MySQL
Reporting & Visualization	Tableau, Power BI, Quick Sight
Big Data Technologies	Databricks, Hadoop, Hive, Apache Spark, MapReduce
CI/CD Tools	Jenkins, GitHub
Operating Systems	Windows, Linux, Unix, AWS EC2, MacOS

Professional Experience:

Client: Fontainebleau Las Vegas, Irving, TX

Mar 2025 – Present

Role: Data Engineer

Responsibilities:

- Designed and implemented end-to-end ETL pipelines using **AWS Glue** and **PySpark**, enabling seamless ingestion, transformation, and loading into **Amazon Redshift** for downstream analytics.
- Automated decryption, extraction, and transformation tasks using containerized services in **AWS ECS** with **ECR**, replacing manual **EC2** workflows and boosting daily pipeline throughput by 30%.
- Orchestrated 30+ batch workflows with **Apache Airflow**, ensuring 99% on-time delivery of datasets for marketing analytics and reporting.
- Provisioned AWS resources (**S3 buckets, SQS queues, IAM roles**) via **Terraform**, enabling 100% reproducible infrastructure deployments and reducing manual configuration errors.
- Optimized complex **SQL** queries, improving query execution speed by 25% across reporting pipelines.
- Implemented event-driven triggers using **AWS Lambda** Functions to automate **Python** scripts, reducing infrastructure management overhead.
- Automated data processing workflows with **Airflow** operators, ensuring timely execution and delivery of critical processes within **AWS**.
- Utilized **Docker** for containerizing **Python** scripts, enhancing their portability and reproducibility.
- Processed terabytes of **structured/unstructured data (CSV, JSON, Parquet)** in **S3** with **PySpark**, performing advanced transformations for downstream analytics.

Environment: Apache Airflow, Python, PySpark, Docker, AWS (Redshift, EC2, ECR, Glue, Lambda, IAM Roles), Terraform, Git, Jira.

Client: Capital One, McLean, VA

Jul 2024 – Nov 2024

Role: Software Engineer - Data

Responsibilities:

- Engineered and implemented **IAM roles** and **policies** for AWS services such as **S3, Glue, Step Functions**, and **EventBridge** using infrastructure as code (Bogie file), improving security and compliance across the data migration project.
- Developed and **optimized 20+ AWS Glue jobs** and **Lambda** functions using Python, automating **ETL processes** and reducing manual intervention by **80%**.
- Designed and built scalable data pipelines using **AWS Redshift, S3, and Kinesis**, processing terabytes of data daily, and improving data **throughput by 40%**.
- Created automated pipelines in AWS Code Pipeline to deploy **Docker** containers in **AWS ECS** using **S3**.
- Developed **REST APIs** for accessing and transforming analytics data, enabling seamless integration with external systems.
- Integrated **CloudWatch logs** with **Splunk** using **Python** scripts, enabling real-time monitoring and reducing incident response times by 25%.
- Automated code review and **CI/CD** processes using Git hooks and integration with GitHub Actions and **Jenkins** for enhanced code quality.
- Collaborated with **cross-functional teams** to ensure seamless data migration and transformation, resulting in a **15% reduction** in project delivery time and a **20% improvement** in data accessibility.

Environment: SQL, Python, PySpark, AWS (Glue, Step Functions, EventBridge, Athena, Redshift, S3, Lambda, IAM Roles), Git, PostgreSQL, Docker, Jira.

Client: Verizon, Irving, TX

Jul 2023 – Jun 2024

Role: Data Engineer

Responsibilities:

- Delivered a major data migration project on **AWS** from a Teradata data warehouse to **Snowflake**, achieving 99.5% data transfer success rate.
- Employed **PySpark** on **AWS EMR** clusters for distributed data processing, improving scalability and reducing processing time.
- Developed **Python** scripts for API extraction, web scraping, and automation, processing 1TB+ of data daily and reducing manual extraction time by 20 hours per week.
- Deployed **Apache Spark** streaming API in a Big Data cluster environment for real-time data processing, achieving an average processing latency of under 500 milliseconds and handling data throughput of over 10,000 records per second.
- Integrated **Amazon Kinesis** and **Apache Kafka** for real-time event streaming and used **Databricks** for real-time analytics, reducing latency by 50%.
- Automated scripts and workflow using **Apache Airflow** and shell scripting to ensure daily production execution.
- Designed and executed transformations with **Spark SQL** and DataFrames API, loading structured and semi-structured data from **MySQL** into Spark clusters.
- Built ETL pipelines with **AWS Glue** to load large datasets into the data warehouse, improving reliability and scalability of ingestion.
- Optimized **Snowflake** environment to remove redundancy and integrated **Kafka** streams into **HDFS** for real-time processing.
- Used **Kubernetes** to orchestrate deployment, scaling, and management of **Docker** containers.
- Created **Tableau** dashboards and reports, managed user permissions, scheduled report refreshes, and maintained high availability on Tableau Server for 100+ users.

Environment: Apache Airflow, Kafka, Spark, MapReduce, Hadoop, Python, Snowflake, Databricks, PySpark, Docker, Kubernetes, AWS, EMR, DynamoDB, CI/CD, Tableau, Teradata, Agile.

Client: Panasonic, Peachtree City, GA

Jan 2020 – Jul 2022

Role: Data Engineer

Responsibilities:

- Designed and deployed automated ETL pipelines using **Azure Databricks** and **Spark SQL**, enhancing market mix modeling and reducing data processing time for sales analytics by 40%.
- Engineered and optimized ETL applications within **Azure Databricks**, improving data extraction and transformation efficiency by 30%, specifically targeting sales data to support strategic decision-making.
- Implemented **Azure Data Factory** for streamlined data extraction from over 20 diverse sources, minimizing errors and enhancing data quality for reliable analysis.
- Executed comprehensive data solutions on **Azure**, leveraging **Data Factory**, **Databricks**, and **Informatica** for advanced processing, governance, and real-time analytics, maintaining a 95% success rate in project implementations.
- Instituted rigorous data quality checks and governance protocols within **Informatica**, ensuring high compliance and accuracy standards for sensitive cancer medicine sales data.
- Designed and maintained scalable data workflows using **Azure Data Factory**, **Event Hubs**, and **Synapse Analytics**, managing over 100TB of data with high efficiency and scalability.
- Developed and monitored systems and processes for data quality assurance, facilitating the integrity and timeliness of data in the company's data lake.
- Designed and developed **Power BI** dashboards and reports, incorporating **DAX** calculations, parameters, measures, hierarchies, and data modeling to deliver actionable business insights.

Environment: Azure (Databricks, Data Factory, Data Lake Storage, Synapse Analytics, Event Hubs), Informatica, Spark, Python, HDFS, Power BI, XML, JSON, Hive, SQL, Agile.

Client: Xoom Works Inc, Hyderabad, India

Jul 2018 – Dec 2019

Role: Data Engineer

Responsibilities:

- Involved in Analysis, Design, System architectural design, process interfaces design, and documentation.
- Developed various **spark applications** using **Scala** to perform various enrichments of user behavioral data (click stream data) merged with user profile data.
- Built production-ready **Spark** solutions using **RDDs**, **DataFrames**, **Spark SQL**, and **Spark Streaming**, improving real-time data processing capabilities.
- Implemented **Slowly Changing Dimensions** (SCD) and other ETL processes using **Informatica**, ensuring historical data was maintained accurately in the data warehouse.
- Created **Tableau** dashboards and visualizations (**charts, graphs, maps, storytelling**) to deliver actionable insights to business stakeholders.
- Developed **MapReduce** jobs in **Pig** and **Hive** for large-scale data cleaning and preprocessing.
- Collaborated in Agile/Scrum teams, contributing to sprint planning, daily stand-ups, and delivery of data solutions.

Environment: Spark, Scala, Hadoop, Python, AWS, Informatica, MapReduce, Pig, ETL, HDFS, Hive, Tableau, SQL.

Certifications: Microsoft Certified: Azure Fundamentals certification (Certification Number: 5B6B4C-787RD3) in Jun 2023

Education: Master of Science in Computer Science, **Texas Tech University**, Lubbock, Texas (Aug 2022 – Dec 2023)